Proceedings of the 6th International Conference on Data Intelligence and Cognitive Informatics (ICDICI-2025)
IEEE Xplore Part Number: CFP25VL6-ART; ISBN: 979-8-3315-0313-0

Enhanced Brain-Computer Interface for EEG Motor Imagery using DC-GCN and Evolutionary Optimization

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2025 6th International Conference on Data Intelligence and Cognitive Informatics (ICDICI) | 979-8-3315-0313-0/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICDICI66477.2025.11135301

Abstract--

Brain-Computer Interface (BCI) systems rely heavily on decoding brain signals for a variety of applications, including motor imagery categorisation. By evading traditional neuromuscular routes, BCI systems enable direct brain-to-device communication. They use electroencephalogram (EEG) and other brain signal detection and interpretation tools to control computers, prosthetics, and other assistive technology. Brain-computer interfaces (BCIs) hold enormous potential for revolutionising neuro-rehabilitation, paralysis communication, and video games. Machine learning and signal processing are advancing at a rapid pace, becoming more accurate and useful. This study introduces a classification framework based on a Densely Connected Graph Convolutional Network (DC-GCN) optimised with Chaotic Sea-Horse Optimiser (CSHO) and BCI Competition IV Dataset 2b. First, EEG signals from C3, Cz, and C4 electrodes are pre-processed using bandpass filtering and Common Average Referencing. features are then extracted using Fast Fourier Transform (FFT). To make efficient use of DC-GCN for classification, extracted features are converted to a one-dimensional format. On average, proposed technique achieves a 92.53% F1-score, 93.29% recall, 92.47% accuracy, and 91.29% precision across all participants, indicating improved performance. DC-GCN + CSHO outperforms other classifiers, including 1D-CNN, RNN, LSTM, DBN, SVM, and ELM, across board. method's ability to capture complex spatial correlations in EEG signals is demonstrated by its higher accuracy than LSTM range of 91.00% and ELM as a89.20%. This method is ideal for real-time BCI applications because it combines graph convolutional networks and evolutionary optimisation, increasing classification robustness. Future research aims to improve this model so that it can be used in assistive neuroprosthetics as well as tasks involving motor imagery across multiple classes. According to findings, DC-GCN + CSHO is an excellent choice for next-generation BCI systems because it achieves ideal balance of precision and computing efficiency.

Keywords: Brain-Computer Interface, Densely Connected Graph Convolutional Network, Chaotic Sea-Horse Optimizer, Motor Imagery, Feature Extraction, Fast Fourier Transform, Deep Learning.

I. INTRODUCTION

BCIs, or brain-computer interfaces, connect brain directly to outside world [1]. BCIs function as closed-loop systems by capturing user's brain activity, analysing it to determine their intentions, and providing feedback by controlling an external device. Electroencephalography (EEG) is a popular non-invasive technique for recording brain waves due to its portability, low cost, and high temporal resolution. Electroencephalograms (EEGs) use electrodes on scalp to record variations in electrical fields generated by brain activity in cerebral cortex [2]. signal captured by each electrode characterises combined activity of millions of neurones, which limits EEG's spatial resolution. Because of their low signal-to-noise ratio (SNR), electroencephalogram (EEG) signals are susceptible to noise introduced by external electrical sources and other physiological processes [3].

BCIs employ a variety of paradigms to elicit identifiable brain patterns in EEG, which aids in decoding and allows for meaningful feedback. This category includes motor imagery (MI), which allows one to voluntarily practise movements without actually performing them [4]. MI stimulates primary motor cortex and related motor areas, mirroring neuronal activity patterns observed during motor execution (ME) [5]. In rehabilitative settings, this overlap of brain circuits is especially beneficial. People with stroke-related motor deficits can improve their skills with BCIs that use magnetic fields (MI) and are integrated into closed-loop systems [6]. Using a variety of machine learning (ML) approaches requires deciphering user intents from brain patterns elicited during tasks. It is common practice to use common spatial patterns (CSP). This method extracts valuable information from EEG data. A linear classifier, such as linear discriminant analysis (LDA), is then used to categorise these characteristics. These conventional approaches have had some success in past, but they are still susceptible to variation between sessions and subjects [7]. Another significant issue with MI-based BCIs is inability to accurately decode EEG data. BCI inefficiency refers to problem that affects 10%-50% of users in MI-based BCI applications. Users frequently fail to achieve effective BCI control due to this inefficiency, which is typically defined as failing to meet 70% accuracy threshold in binary MI tasks [8]. Users who are unable to control application in a rehabilitation setting do not receive important feedback for regaining lost functions. In addition to wasting time, this leads to frustration and a gradual loss of motivation, both of which are necessary for improvement [9]. main reasons BCIs do not work as well as they could are previously mentioned EEG issues, classification phase issues, users' lack of experience with abstract tasks, and a lack of motivation during long periods of skill learning [10]. Deep learning (DL) approaches have recently been focus of extensive research as a means of overcoming these challenges; these methods outperform more traditional ML approaches in terms of MI decoding. DL improves ability to transfer learning (TL) between datasets, sessions, and users, as well as interpretation and extraction of EEG features [11]. As an example, TL enables reuse of MI data from other users to train models capable of performing BCI calibration run.

Calibration is another reason why BCIs are ineffective in traditional MI-based BCI applications [12]. Users are typically not given feedback on their performance while being asked to learn MI tasks during calibration. When a user is new to the MI task and has yet to see any results from their efforts, this becomes significantly more difficult [13]. This can result in a self-perpetuating loop in which an inexperienced user of a brain-computer interface (BCI) is unable to learn how to modulate their own brain activity because they did not provide adequate calibration to provide relevant feedback. Because EEG data can vary both within and between subjects, the ML models used for MI must be recalibrated at each session [14]. This is why TL between users has been the focus of some DL-based classification methods research for MI-based BCIs, with the goal of eliminating the need for calibration runs. However, there are some issues with training on other users' MI data [15]. For example, participants may not have completed the task consistently, which could explain some of the inter-subject variability. Furthermore, it is unable to perform external verification of task performance during data collection. Using MI data to train a deep learning model is a promising solution to this problem. This method makes use of DL's TL capabilities while also leveraging the shared features of MI and ME [16]. In most cases, inter-subject categorisation works better with ME than with MI because the brain patterns induced by ME are more consistent across users. Furthermore, for BCI-based rehabilitation applications, it may be preferable to teach users using a DL model based on these more stable patterns of ME rather than patterns that are variable and diffused from other users' MI. Despite these advantages, direct inter-task TL has not yet been studied [17]. This research introduces a new approach to BCI classification that uses the Chaotic Sea-Horse Optimizer (CSHO) in conjunction with Densely Connected Graph Convolutional Networks (DC-GCN) to improve the categorization of EEG signals pertaining to motor imagery. Important findings from this study include:

- Introducing DC-GCN to effectively model the spatial besides temporal dependencies in EEG signals, improving classification accuracy.
- Implementing Chaotic Sea-Horse Optimization (CSHO) to fine-tune hyperparameters, enhancing the robustness of feature learning.
- Employing Fast Fourier Transform (FFT) on C3, Cz, and C4 electrodes to extract meaningful frequency-domain features, reducing computational complexity.
- Achieving a 92.47% accuracy, surpassing existing classifiers such as 1D-CNN, RNN, LSTM, DBN, SVM, and ELM, demonstrating the effectiveness of the proposed approach.
- Designing a framework that can be adapted for multi-class classification, cross-subject adaptability, real-time processing, and deployment on wearable BCI devices.

The rest of the paper is organized as surveys: Section 2 mentions related works; Section 3 provides the proposed methodology in detailed; Section 4 discuss the result analysis and finally, the conclusion is made at Section 5.

II. RELATED WORKS

The Lightweight Brain-Computer Interface (LGL-BCI) is a prototype developed by Lu et al., [18]. It employs our specialized geometric deep learning architecture to quickly infer models without compromising accuracy. In order to accommodate the ever-changing EEG signal, LGL-BCI includes a channel selection module that uses a feature decomposition approach to lower dimensionality of a symmetric positive definite matrix. Meanwhile, inference speed is enhanced by an integrated lossless transformation. To used two publicly available EEG datasets and two real-world EEG equipment to test how well our method worked. With an accuracy of 82.54%, LGL-BCI significantly outperformed state-of-the-art method, which achieved 62.22%. computational economy of LGL-BCI is further demonstrated by fact that it needs fewer parameters (64.9K vs. 183.7K). These results show that geometric deep learning is feasible and robust in applications involving motor-imagery brain-computer interfaces, and they also highlight better accuracy and computational efficiency of LGL-BCI.

To increase classification accuracy of motor components of BCI systems—Mallat et al., [19] introduced a novel technique that leverages collective synergy of five convolutional neural network (CNN) representations. By outperforming current state-of-the-art methods that use multiple CNN models, our method achieves an impressive accuracy of 79.44% on BCI Competition IV 2a dataset. This development shows great potential for improving efficiency and adaptability of BCIs in various practical uses, such as neurorehabilitation and assistive technology, and thereby offering strong assistance to those who have motor impairments.

goal of innovative and effective hybrid BCI model developed by Zhao et al., [20] was to build a matching dataset and improve accuracy and range of MI by integrating EEG and electromyogram (EMG) signals. Using following steps: converting classification problem into a regression problem and constructing an ensemble learning model called MOBF; applying bilinear submanifold learning (BSML) and minimum distance to bilinear submanifold mean (MDSM) algorithms to EEG signals; and implementing sliding windows, L1 regularization, and support vector machine (SVM) algorithm to EMG signals. In addition, model's presentation was evaluated by extensive experiments using both publicly accessible datasets and datasets that were created in-house using Leave-one-out cross-validation (LOOCV). In comparison to state-of-the-art (SOTA) methods, suggested MOBF model improves MI accuracy by 2.4% on home-made dataset and 1.1% on public dataset, according to results. It is worth mentioning that model not only achieved highest accuracy for muscle tiredness detection but also across multiple EEG frequency bands.

A full MI dataset was gathered by Yang et al., [21] from the 2019 World Robot Conference Contest-BCI Robot Contest. Across three recording sessions, to obtained EEG data from sixty-two healthy individuals. There are two paradigms in this experiment: (1) tasks that require both the left and right hand to grasp; (2) tasks that need both the left and right hand to grip; and (3) tasks that require the foot to hook. Included in the dataset are both raw and pre-processed data sets. While deepConvNet attained a classification accuracy of 76.30% with three-class data, EEGNet averaged 85.32 percent with two-class data. The dataset can be reused by different researchers as needed. In particular, to are hoping that this dataset will help researchers tackle cross-session and cross-subject problems using MI-BCI.

In order to improve the BCI application, Bhalerao et al. [22] introduced a classification framework for MI tasks using recorded EEG signals that is based on Fourier-Bessel series expansion (FBSE). When compared to the Fourier spectrum, the spectral resolution of the FBSE spectrum is superior for nonstationary signals. It gives a positive-frequency representation of real signals and does away with the need for a window function to get the signal's spectrum. In order to isolate rhythms specific to MI from EEG signals, the FBSE decomposition method is useful because of its novel and condensed spectrum representation. In order to better categorize the tasks, the suggested study seeks to improve rhythm separation and feature extraction. The EEG rhythms have been used to extract multidomain properties, such as band power and Hjorth parameters. Random forest, ensemble KNN, support vector machine, linear discrimination analysis are the five separate classifiers that have been used in the 10-fold cross-validation process. Incorporating the student t ranking strategy into the established automated system has further enhanced its performance. When compared, the suggested classification framework achieves an accuracy of 86.12% on features collected from FBSE based rhythms using the ensemble KNN classifier.

III. PROPOSED METHODOLOGY

Analyzing EEG motor imagery signals efficiently presents a number of obstacles for traditional Brain-Computer Interface (BCI) classification algorithms. One problem with 1D-CNNs is that they have trouble capturing dependencies over large distances. Another problem with RNNs and LSTM networks is that they have high processing costs and vanishing gradients. Extreme Learning Machines demonstrate potential, however they can't handle variances among subjects. The classification accuracy of Support Vector Machines (SVM) is lowered when dealing with high-dimensional EEG feature spaces.

To overcome these obstacles, the DC-GCN + CSHO system models the spatial and temporal connections in EEG data using graph-based deep learning. Keeping signal correlations intact, DC-GCN learns feature associations across EEG channels more effectively

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than CNNs and RNNs. To further improve model generalization and optimization, which in turn improves classification accuracy and stability, the Chaotic Sea-Horse Optimizer (CSHO) has been integrated. In addition, as demonstrated in Figure 1, Fast Fourier Transform (FFT) guarantees efficient frequency-domain feature extraction while lowering computing complexity.

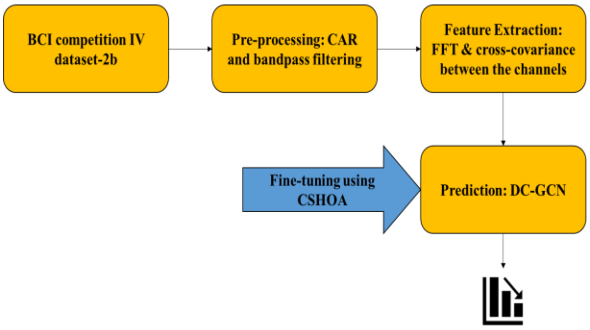

Figure 1: Workflow of the Research Methodology

A. Dataset Description

Dataset 2b from BCI competition IV is utilized in this investigation. Nine subjects' electrooculograms (EOGs) and electroencephalograms (EEGs) are included in the recordings [23]. All of the subjects are right-handed and have normal vision. Everyone sat in a cozy armchair, staring at a screen that was precisely one meter away. The nine individuals are split into two groups besides are then given MI tasks that involve visualizing movements to the left and right. Over the course of five sessions, EEG signals were obtained from every participant. Although all three sessions included comments, the first two did not. To initiate the first data collection session, to utilized the preoccupied cross and alert tone to signal the commencement of data collection. After seeing a 1.25 s visual cue, participants choose between two motor imagining activities. Researchers used a passband frequency range of 50–100 Hz to sample EEG signals at 250 Hz in the experimental dataset.

B. Preparation of data

Dataset 2b from BCI competition IV, which contains information from the first three sessions of EEG signals, was analyzed in the proposed study. Three of the six channels in the sample are electroencephalograms (EEGs), and the other three are heart rate variability (HRV). For the purpose of MI task prediction, to omit the EOG channels and utilized only three EEG channels. Since EEG data contains low-frequency noise, its analysis is complicated. Consequently, to use the CAR method to recover the raw EEG signal-to-noise ratio. Produced by implementing CAR. After that, it goes via a bandpass filter that operates between 8 and 30 Hz for additional processing. Figure 2 displays the three different versions of Subject1's EEG data: the raw signal, the CAR-affected signal, and the bandpass-filtered signal (C3, Cz, and C4 channels). To properly manage the risk of overfitting, to also included a sliding window. From each signal that has been CAR-filtered, to take a 4-second signal (3s to 7s) component. Afterwards, for a maximum of eleven signal segments, a two-second sliding window is employed, with a shift of 0.1 s. Lastly, eleven 2-second signal segments are generated for every EEG session. In order to test the suggested model, this procedure expands the original dataset.

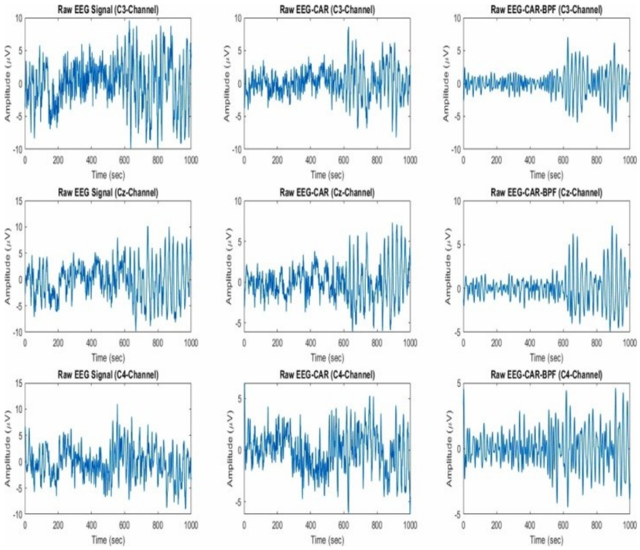

Figure 2: Sample of pre-processed signal

C. Feature Extraction

At the feature extraction step, to aimed to compare the system's performance for motor imaginary recognition using time-domain and frequency-domain feature extraction methods.

The use of numerous electrodes in an EEG typically results in a time series with many dimensions. As a general rule, in order to compress EEG data, one must first perform a feature selection-based data compression step to extract the crucial information from the signals [24] or to minimize the number of channels according to their informative relevance in respect to the system goal. In order to capture the electrode variability, a novel method for lowering the data dimension of EEG signals was introduced. This method involved computing the cross-covariance domain. To extended our feature extraction method to the frequency domain and applied it in this investigation as well.

This study described the cross-covariance of the C3, Cz, and C4 channels as:

$$Cov\left(X^{c3}(t), X^{cz}(t), X^{c4}(t)\right) = E\left[[X^{c3}(t) - E(X^{c3}(t))][X^{cz}(t) - E(X^{cz}(t)][X^{c4}(t) - E(X^{c4}(t))]\right] \quad (1)$$

where $X^{c3}(t)$ characterises the EEG signal learnt for channel c3, $X^{cz}(t)$ is the EEG signal acquired for channel cz, and $X^{c4}(t)$ is acquired for channel c4 channel. $E[X^{ch}(t)]$ characterizes the expected value (where ch corresponds to the explicit channel c3, cz and c4) and is computed as:

$$E\left(X^{ch}(t)\right) = \frac{1}{W}\sum_{i=0}^{W-1} x_i^{ch} \ (2)$$

The feature computation window dimension is represented by the W value in Equation (2).

Using the Fast Fourier Transform (FFT), the second feature extraction approach examined in this research transforms time-domain EEG signal series into frequency-domain signals. To separate the signal into its component sine and cosine waves, one can apply the Fourier transform.

In order to calculate the channel's FFT, the following:

$$FX^{ch}(f) = \sum_{t=0}^{n-1} X_t^{ch} e^{-\frac{j2x\,ft}{n}} \ (3)$$

where X_t^{ch} represents the EEG acquired for channel ch.

To calculated the cross-covariance between the Fourier transform of the channels after obtaining the signals corresponding to required channels using Equation (3).

A 1D feature matrix for a 2s window period with 128 dimensions, one for the time domain and one for the frequency domain. The DC-GCN classification process is then initiated with the features that have been extracted.

D. Densely Connected Graph Convolution

Graph Convolution

The input, hidden, and output layers comprise the multi-layer GCN neural network. It uses the adjacency matrix to learn node characteristics, word co-occurrence vectors as nodes, and word co-occurrence relationships as edges. With node feature matrix in particular$X = [x_1, x_2, x_3, \dots, x_n]$ and adjacency matrix A. diagonal elements of A to be set to 1. In addition, to also introduce a degree matrix $D \in R^{n*n}$, where each element D_{ii} in D is:

$$D_{ii} = \sum_j A_{ij} \ (4)$$

Therefore, for a GCN with one layer, the learning development of node features is:

$$L^{(1)} = relu\left(\tilde{A}XW_0\right) \ (5)$$

where $\tilde{A} = D^{-1/2}AD^{-1/2,}, W_0$is a trainable limit.

Dense Connection

As seen in Section 3.3.1, a single-layer GCN is limited to taking the properties of its immediate neighbors. In contrast, a multi-layer GCN can acquire the crucial features of a multi-node graph by capturing the attributes of faraway nodes via frequent surrounding nodes. This formula characterises multi-layer GCN stacking.:

$$L^{(m)} = relu\left(\tilde{A}L^{(m-1)}W_{m-1}\right) \ (6)$$

where m is the number of layers of GCN.

On the other hand, when the network gets deeper, the shallow characteristics might disappear. Direct combination of shallow and deep features is possible with dense connections. Our densely linked GCN network (DC-GCN) takes architectural cues from DenseNet and uses feature multiplexing in shallow GCN layers to improve node feature capture across longer distances.

In light of this, DC-GCN's m-th layer broadcast becomes:

$$L^{(m)} = relu\left(\tilde{A}\left[L^{(1)} \oplus L^{(2)} \oplus \ , \dots, L^{(m-1)} \oplus\right] W_{m-1}\right) \ (7)$$

where splicing is denoted by $\oplus$. To deviate from the conventional tactic by retaining output from each layer and generating the final output from intermediate characteristics at different scales.

$$h_i = L_i^{(1)} \oplus L_i^{(2)} \oplus \dots \oplus L_i^{(m)} \ (8)$$

$$H = [h_1, h_2, \dots h_n] \ (9)$$

where hi is the i −th word node. Therefore, finished this layer, to get last feature matrix $H \in R^{n*(k*m)}$, where k is GCN.

Attention Mechanism

The retrieved features must be joint into a classified. For the feature matrix $H = [h_1, h_2, h_3, \dots, h_n]$, Not all features are job. Consequently, to build the attention module to emphasize their importance. An attention score a_i is calculated for each feature hi; a_i signifies the feature's significance in relation to classification task:

$$a_i = \frac{exp(e_i^T u_s)}{\sum_i exp(e_i^T u_s)} \ (10)$$

$$e_i = relu(W_s h_i + b) \ (11)$$

where W_s and us are trainable bias term.

Lastly, to multiply score on care $\alpha = [a_1, a_2, a_3, \dots, a_n] \in R^n$ and the feature matrix H. Sum finished illustration V:

$$V = \sum_{i \in n} a_i h_i \ (12)$$

Through this layer, to become final picture $V \in R^{km}$.

Classification

A single fully connected layer is utilized by the categorization module. This layer's job is to use the final representation V to determine the distribution. Here is the equation that represents this layer::

$$P = softmax(WV + b) \quad (13)$$

where the probability distribution of the category is represented by P. The trainable parameter is denoted by W, bias term is denoted by b, besides the activation function is softmax.

E. Hyper-parameter tuning using CSHO

The parameters of the DC-GCN model are fine-tuned using CSHO [25], with the fundamental SHO presented first.

Sea-Horse Optimizer (SHO)

The SHO takes its seahorse's natural behaviors, such as swimming, mating, and hunting for food. The SHO algorithm's metaheuristic approach is based on the idea of exploration and exploitation, and it's built to mimic the social as they hunt for food. When both parts are complete, the breeding process is complete. Detailed modeling of the SHO technique is possible in the following way::

$$Sh = \begin{bmatrix} x_{1,i} & \cdots & \cdots & x_{1,Dim-1} & x_{1,Dim} \\ x_{2,i} & \cdots & & \cdots\ \cdots & x_{2,dim} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ x_{N,i} & \cdots & \cdots & x_{N,Dim-1} & x_{N,Dim} \end{bmatrix} \quad (14)$$

$$Sh_{ij} = Rand \times \left(UB_j - LB_j\right) + LB_j \quad (15)$$

$$Sh_{elite} = argmin(f(X_i)) \quad (16)$$

Where $\boldsymbol{X}im$ denotes the variable's dimension and N stands for the population's size. The random results from each solution are represented by SB and UB, respectively, and they represent the upper and lower limits. $Rand$ is a symbol for a random value that can take on values between 0 and 1. The $Shelite$ represents elite individuals who have at least a basic level of fitness. SHO takes on the behaviors of seahorses, including swimming, foraging, and mating.

F. Seahorse Movement Behavior

The way seahorses move is based on distribution. finding the sweet spot between discovery and exploitation by comparing two case studies with a 0.

Case 1: In a spiraling motion, the agent approaches the X_elite, continuously adjusting the rotation angle to expand the local key zone. The mathematical formulation for Case 1 is as follows:

$$X_n(t+1) = X_i(t) + Levy(\lambda)\left(X_{elite}(t) - X_i(t)\right) \times x \times y \times z + X_{elite}(t)) \quad (17)$$

$$\theta = rand \times 2\pi \quad (18)$$

$$x = \times\ cost(\theta) \quad (19)$$

$$y = \rho \times \sin(\theta) \quad (20)$$

$$z = \rho \times \theta \quad (21)$$

$$\rho = \mu \times e^{\theta v} \quad (22)$$

$$Levy(\lambda) = s \times \frac{w \times \sigma}{|K|^{\frac{1}{\lambda}}} \quad (23)$$

In this case, ρ is a randomly chosen integer between 0 and 2, and the length of the rod is defined by the constant u (default=0.05) and a (default=0.05). The values of k and w are chance integers between zero and one. A constant of 0.01 is denoted by s.

Case 2: on order to improve its traverse, seahorses use a Brownian motion that imitates the motion length of other seahorses when swimming on ocean waves. Here is one way to put it::

$$X_n(t+1) = X_i(t) + rand \times l \times \beta_i \times (X_i(t) - \beta_i \times X_{elite}(t)) \quad (24)$$

$$\beta_i = \frac{1}{\sqrt{2\pi}} exp\left(-\frac{x^2}{2}\right) \quad (25)$$

$$X_n(t+1) = \begin{cases} X_i(t) + Levy(\lambda)\left(\left(X_{elite}(t) - X_i(t)\right)\right) \times x \times y \times z + X_{elite}(t)) & if\ r_1 > 0 \\ X_n(t+1) = X_i(t) + rand \times l \times \beta_i \times (X_i(t) - \beta_i \times X_{elite}(t)) & if\ r_1 \leq 0 \end{cases} \quad (26)$$

where βi for Brownian motion, where is the random walk coefficient. Default value for l is 0.5, which is a constant. $r1$ represents an arbitrary number.

G. Seahorse Foraging Behavior

There are two possible outcomes when seahorses search for food: success and failure. A value of $f2 > 0.1$ indicates that the condition has been successfully set. In this scenario, the seahorse outpaces its prey in terms of speed. Conversely, a failure condition arises in cases where the reaction differs. What follows is a formulation of the conditions under which seahorses succeed or fail in their search for food:

$$X_n(t+1) = \begin{cases} a \times \left(X_{elite}(t) - rand \times X_{new}(t)\right) + (1-a) \times X_{elite}(t)) & if\ r_2 > 0 \\ (1-a) \times (\left(X_{new}(t) - rand \times X_{elite}(t) + (a) \times X_{new}(t)\right) & if\ r_2 \leq 0 \end{cases} \quad (27)$$

$$a = (1 - \frac{t}{T})^{\frac{2t}{T}} \quad (28)$$

where $Xnew$ new is novel position of seahorse. $r2$ is a random sum [0, 1]. T is dangerous iteration.

H. Seahorse Breeding Behavior

There are two sex groups in seahorses, the males and the females, and their relative proportions are exactly equal (50/50).

$$Father = X_{sort}\left(1:\frac{pop}{2}\right) \quad (29)$$

$$Mother = X_{sort}\left(\frac{pop}{2}+1:pop\right) \quad (30)$$

Sorted X_{sort} the result is returned in ascending order of values. Both $Fabher$ and $Mobher$ were selected at random. Every couple has a single offspring according to the SHO algorithm.

$$X_i = (1-r_3)X_{mother} + r_3X_{father} \quad (31)$$

Where $r3$ is a random sum [0, 1].

The Novel Chaotic Sea-Horse Optimizer (CSHO)

The optimization of algorithms has been the subject of numerous publications that have employed various kinds of chaotic maps. Chaotic maps are characterized by their dynamic nature and statistics that are based on unpredictability. So that the results vary depending on how the settings are tweaked. Equation (18) is substituted with the logistic type chaotic map in this article. The logistic category chaotic map's mathematical equation looks like this:

$$ylog_{(i+1)} = a \times ylog_{(i)}(1-ylog_{(i)}) \quad (32)$$

Where a equals 4. The resulting variable is the chaotic map, which can take values between 0 and 1. Therefore, when to do this, Equation (18) becomes Equation (33).:

$$\theta = ylog \times 2\pi \quad (33)$$

IV. RESULTS AND DISCUSSION

The proposed model is evaluated on workstation with configuration as: An Intel (R) Xeon (R) W-2125 4.01 GHz central processing unit (CPU), a GeForce (R) GTX1080TI GPU, and 64 GB of RAM using MATLAB R2022b [26]. Figure 3 mentions the confusion matrix for all the subjects.

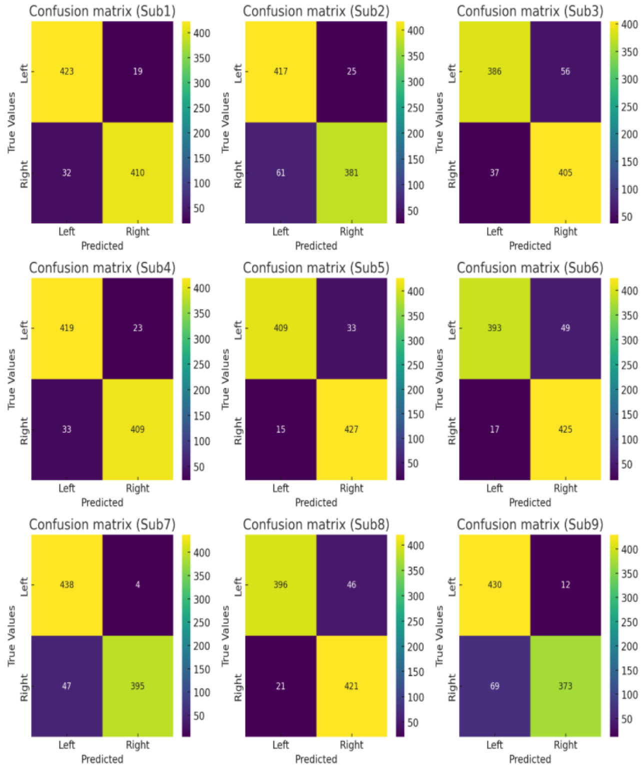

Figure 3: Confusion matrixes of 9 different subjects.

Analysis of proposed model on Dissimilar Subjects

Table 1 and Figure 4 presents the performance study of proposed classical with different subjects in terms of various metrics.

Table 1. Nine participants' accuracy, precision, recall besides F1-score comparison.

Subject	**Accuracy**	**Precision**	**Recall**	**F1-score**
Sub1	0.9423	0.9570	0.9297	0.9431
Sub2	0.9027	0.9434	0.8724	0.9065
Sub3	0.8948	0.8733	0.9125	0.8925
Sub4	0.9367	0.9480	0.9270	0.9374
Sub5	0.9457	0.9253	0.9646	0.9446
Sub6	0.9253	0.8891	0.9585	0.9225
Sub7	0.9423	0.9910	0.9031	0.9450
Sub8	0.9242	0.8959	0.9496	0.9220
Sub9	0.9084	0.9729	0.8617	0.9139
Average	**0.9247**	**0.9329**	**0.9199**	**0.9253**

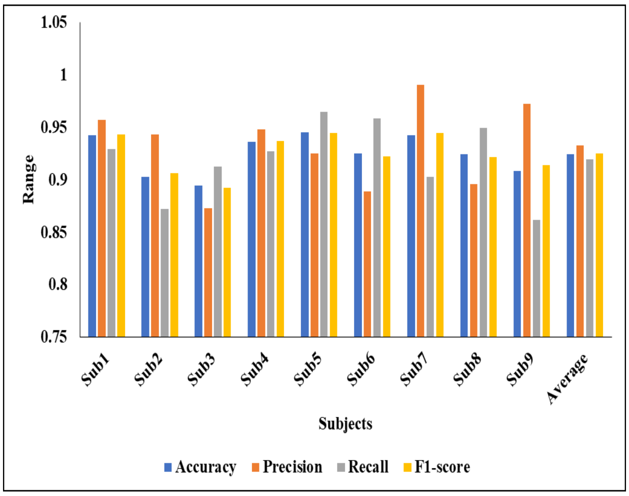

Figure 4: Analysis of proposed model on nine subjects

A comparative analysis of score across nine subjects to evaluate the model's presentation on individual participants. The highest accuracy is achieved by Sub5 (94.57%), closely followed by Sub1 (94.23%) and Sub7 (94.23%). Precision is highest for Sub7 (99.10%), while Sub5 also shows strong recall (96.46%) and F1-score (94.46%). Sub3 reports the lowest accuracy (89.48%) but maintains a balanced performance in recall and F1-score. The overall average performance across all subjects is solid, with 92.47% accuracy, 93.29% precision, 91.99% recall, and a consistent F1-score of 92.53%, indicating the model's robust and generalizable classification ability across different individuals [27].

Comparative analysis of projected prototypical with existing techniques

Table 2 and Figure 5 provides the comparative analysis of planned classical with existing techniques in terms of various metrics.

Table.2. Proposed model compared with various existing systems.

Model	Accuracy	Precision	Recall	F1-score
1D-CNN	0.8900	0.8850	0.8800	0.8825
RNN	0.8650	0.8600	0.8550	0.8575
LSTM	0.9100	0.9050	0.9000	0.9025
DBN	0.8750	0.8700	0.8650	0.8675
SVM	0.8800	0.8750	0.8700	0.8725
ELM	0.8920	0.8870	0.8820	0.8845
DC-GCN + CSHO (Proposed)	**0.9247**	**0.9329**	**0.9199**	**0.9253**

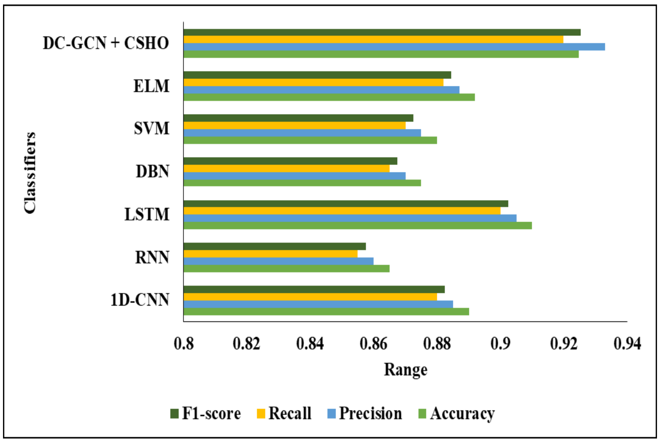

Figure 5: Visual analysis of various models

A comparative evaluation of the proposed DC-GCN + CSHO model against various existing systems based on key performance metrics—score. Among all models, the proposed approach demonstrates the highest accuracy of 92.47%, precision of 93.29%, recall of 91.99%, and F1-score of 92.53%. Traditional models like RNN and DBN show lower presentation, with accuracies of 86.50% and 87.50%, respectively [28]. While models like LSTM and ELM perform relatively better (91.00% and 89.20% accuracy), they still fall short compared to the proposed system. This highlights the superior capability of the proposed DC-GCN + CSHO framework in achieving balanced and high-quality predictions for the given task [29].

V. CONCLUSION

This study introduces an advanced BCI system for EEG-based motor imagery classification, leveraging a DC-GCN optimized with the Chaotic Sea-Horse Optimizer (CSHO). By applying Common Average Referencing (CAR) besides bandpass filtering on C3, Cz, and C4 electrode signals, followed by Fast Fourier Transform (FFT) for feature extraction, the proposed model effectively transforms EEG signals into a one-dimensional format for classification. The results demonstrate a significant upgrading in presentation, achieving an accuracy of 92.47%, precision of 93.29%, recall of 91.99%, and F1-score of 92.53%, outperforming traditional approaches such as 1D-CNN, RNN, LSTM, DBN, SVM, and ELM. The study highlights the effectiveness of graph-based models in capturing complex spatial besides temporal dependencies in EEG signals. The combination of DC-GCN and CSHO enhances classification accuracy while efficiency, making it a viable solution for real-time BCI applications. Future work will focus on expanding to multi-class motor imagery classification, improving cross-subject adaptability, optimizing real-time processing for neuroprosthetics, and integrating hybrid EEG features for enhanced classification robustness. Additionally, deploying the model on wearable EEG devices and edge hardware can further facilitate practical applications in healthcare, rehabilitation, and assistive robotics, making BCI technology more accessible and efficient.

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